THEORY ASSIGNMENT – 3

Submitted By: Harshit Kumar Gautam

University Roll No.: 22615990003

Course: B-TECH (CSE)

Section: NA (L1)

College: GLA University

Submitted To: Mr Gautam Mukherjee

1. What is the difference between frontend and backend JavaScript?

Frontend JavaScript runs in the user's web browser. It makes a webpage interactive, validates forms, responds to clicks, and updates page content. Backend JavaScript runs on a server, commonly with Node.js. It handles requests, business logic, authentication, and database operations.
Example: Displaying a message when a button is clicked is frontend work, while checking login details in a database is backend work.

Code Example:

```
// Frontend
document.getElementById("msg").textContent = "Welcome";

// Backend (Node.js)
const http = require("http");
http.createServer((req, res) => {
res.end("Hello from server");
}).listen(3000);
```

2. What is DOM?

DOM stands for Document Object Model. It represents an HTML document as a tree of objects called nodes. JavaScript uses the DOM to access and change the content, structure, and styles of a webpage.

Code Example:

```
// HTML: <h2 id="title">Old Text</h2>
document.getElementById("title").textContent = "New Text";
```

3. What is an event in JavaScript?

An event is an action detected by the browser, such as a mouse click, key press, form submission, or change in an input field. JavaScript can run a function in response to an event.

Code Example:

```
document.getElementById("btn").addEventListener("click", function () {
alert("Button clicked!");
});
```

4. What is the event loop in JavaScript?

The event loop helps JavaScript handle asynchronous operations. Synchronous code runs first. When the call stack is clear, queued callbacks can run. Promise callbacks (microtasks) are processed before the next task, such as a timer callback.

Code Example:

```
console.log("Start");
setTimeout(() => console.log("Timer"), 0);
console.log("End");

// Output:
// Start
// End
// Timer
```

5. What is hoisting in JavaScript?

Hoisting describes how JavaScript processes declarations before executing code in their scope. Function declarations can generally be called before their written position. A variable declared with var is initialized as undefined, while let and const cannot be accessed before initialization.

Code Example:

```
console.log(x); // undefined
var x = 10;

hello();
function hello() {
console.log("Hello");
}
```

6. What is event bubbling?

Event bubbling occurs when an event starts at the element where it happened and then travels upward through its parent elements. For example, clicking a button inside a div may trigger click handlers on both the button and the div. The method event.stopPropagation() can stop further propagation.

Code Example:

```
document.getElementById("parent").addEventListener("click", () => {
console.log("Parent clicked");
});
document.getElementById("child").addEventListener("click", () => {
console.log("Child clicked");
});
```

7. What is the difference between named, anonymous and arrow functions?

A named function has a name and can be called by that name. An anonymous function has no name of its own and is often assigned to a variable or passed as a callback. An arrow function uses the => syntax and provides a shorter way to write functions. Arrow functions use the surrounding this value.

Code Example:

```
// Named function
function add(a, b) {
return a + b;
}

// Anonymous function
const sub = function(a, b) {
return a - b;
};

// Arrow function
const multiply = (a, b) => a * b;

console.log(add(2, 3)); // 5
console.log(sub(5, 2)); // 3
console.log(multiply(2, 3)); // 6
```

8. Explain for-of and for-in loops.

The for...of loop is used to access values from iterables such as arrays and strings. The for...in loop is used to access enumerable property keys of an object. With arrays, for...in gives index keys, not the actual values.

Code Example:

```
const nums = [10, 20, 30];
for (const value of nums) {
console.log(value);
}

const student = { name: "Aman", age: 19 };
for (const key in student) {
console.log(key, student[key]);
}
```

9. Explain any five array methods.

Array methods are built-in functions used to add, remove, transform, or select array elements. Five common methods are push(), pop(), map(), filter(), and forEach().

Code Example:

```
// 1. push(): adds an item at the end
let a = [1, 2];
a.push(3); // [1, 2, 3]

// 2. pop(): removes the last item
a.pop(); // removes 3

// 3. map(): transforms every item
[1, 2, 3].map(n => n * 2); // [2, 4, 6]

// 4. filter(): selects matching items
[1, 2, 3, 4].filter(n => n % 2 === 0); // [2, 4]

// 5. forEach(): runs a function for each item
["A", "B"].forEach(item => console.log(item));
```

10. What is a callback function?

A callback function is a function passed as an argument to another function. The receiving function calls it later or when a particular action occurs. Callbacks are commonly used in event handling, timers, and asynchronous programming.

Code Example:

```
function greet(name, callback) {
console.log("Hello, " + name);
callback();
}

function welcome() {
console.log("Welcome!");
}

greet("Aman", welcome);

// Output:
// Hello, Aman
// Welcome!
```